A²Agent: Action-Aware Reinforcement Learning for Repository-Level Code Localization Agents

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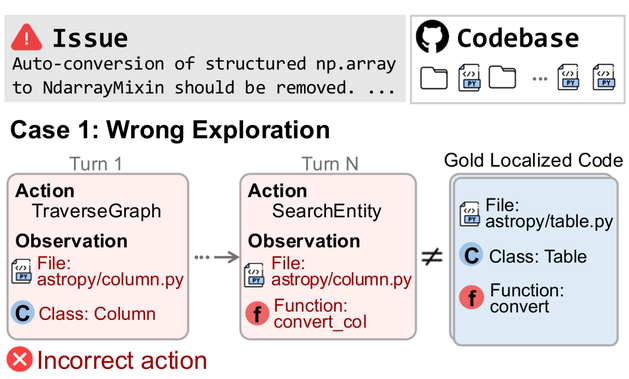
Code Paper Profile

1. Problem Definitions

Background

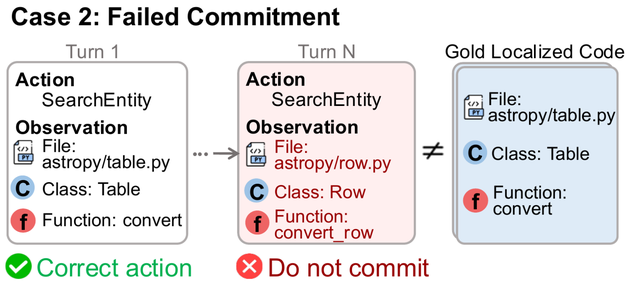
- **Code localization**, finding the code regions to modify for an issue, is a **prerequisite for issue resolution**.
- At the repository level, agents **invoke multiple tools over many turns** to navigate the codebase.
- They are trained by SFT, or by RL where the **reward comes only from the final prediction**.
- With such **sparse trajectory-level signals**, **per-action credit assignment is difficult**.

Problem



- Cannot identify effective actions, causing unnecessary tool calls.

Trajectory-level rewards cannot tell which per-turn actions are effective.



- Gold regions are observed but not committed: 40%+ of baseline failures.

Idea

❶ Per-Turn Reward Sequence : Rewarding both discovery and commitment of gold code regions.

❷ Action-Level Advantage Estimation : Isolating each action's credit by grouping turns that share the same exploration context.

2. Proposed Method

Key Components

A. Reward Design

- **Turn-level reward**: rewards each newly discovered gold file, module, or function, so **partial successes still get a signal**.
- **Terminal reward**: final F1 plus the Discovery-Commit ratio, the **share of explored gold locations that are committed**.

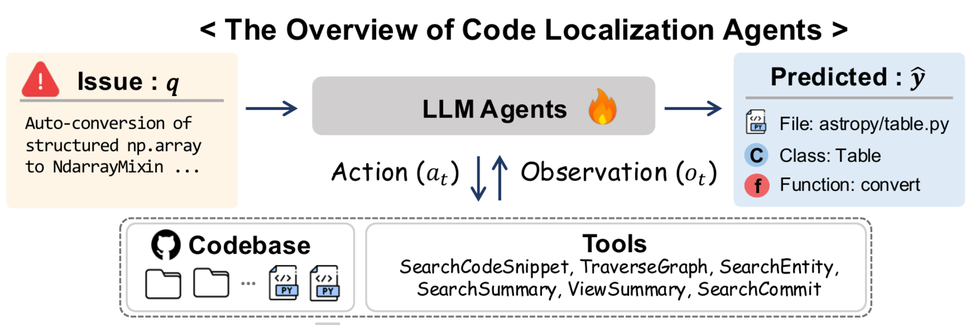
B. Action-Level Advantage Estimation

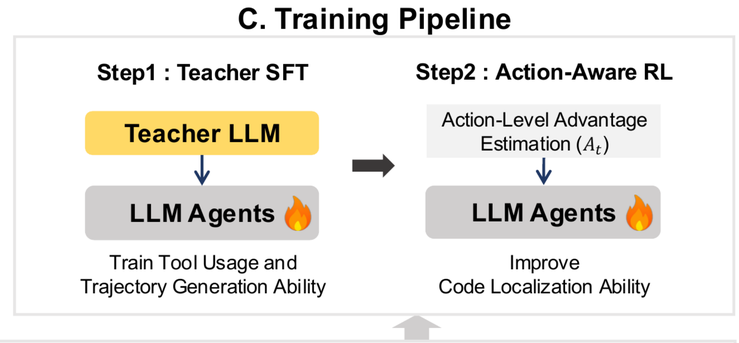
- Across rollouts of the same issue, turns whose preceding actions match in **both tool and arguments** are grouped, and each action is **compared with the alternatives** in its group.
- Several history depths and a trajectory-level anchor give **every turn a stable signal**.

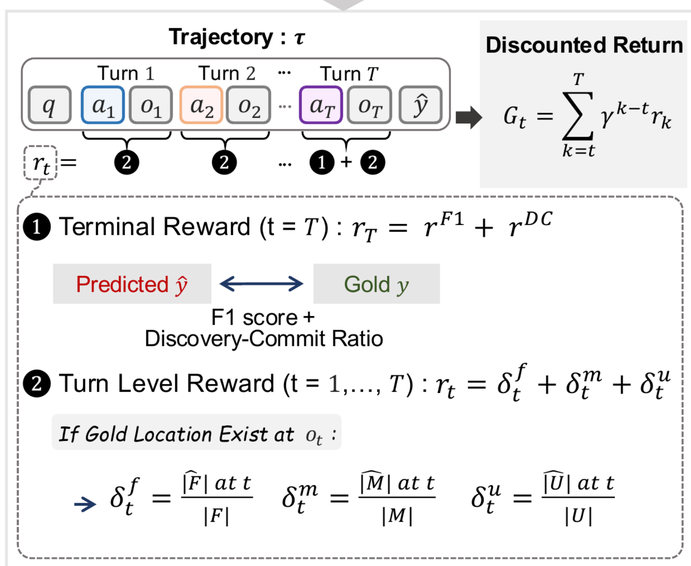
C. Training Pipeline

- **Step 1, Teacher SFT**: builds tool-use and trajectory generation ability.
- **Step 2, Action-Aware RL**: reinforces high-advantage actions.

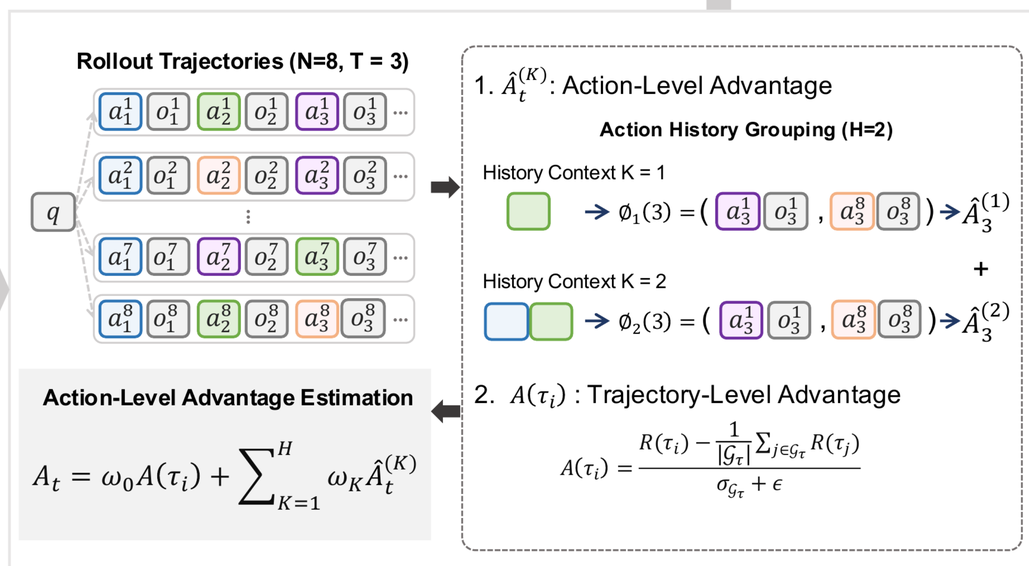
A²Agent Framework







A. Reward Design



B. Action-Level Advantage Estimation

3. Evaluation Results

- A²Agent outperforms all open-source baselines, improving average F1 over SOTA by **1.58% on SWE-Bench Verified** and **8.55% on SWE-Bench Pro**.
- Our **4B model surpasses baselines up to 8× larger**, with the **lowest inference time** among agent-based methods and the **highest resolve rate**.
- **No external API access or Docker setup** is required, so it can be deployed entirely on-premises where source code privacy matters.

Method	LLM	File-level F1	File-level *P*	File-level *R*	Module-level F1	Module-level *P*	Module-level *R*	Function-level F1	Function-level *P*	Function-level *R*
		Closed-Source LLMs								
RepoSearcher	Claude-3.7-Sonnet	32.30	20.24	**89.24**	–	–	–	26.91	18.64	**66.08**
RepoNavigator	GPT-5-Chat	58.88	61.87	58.17	–	–	–	31.17	34.56	30.42
	Claude-3.7-Sonnet	73.01	75.95	72.26	–	–	–	31.72	34.43	31.03
	Claude-Sonnet-4.5	79.94	81.92	80.68	–	–	–	43.62	45.76	43.97
OpenHands-Bash	GPT-5	3.20	3.20	3.20	2.60	2.60	2.60	2.60	2.60	2.60
	Claude-Sonnet-4.5	0.80	0.80	0.80	0.40	0.40	0.40	0.40	0.40	0.40
OpenHands-Bash^{rem}	GPT-5	78.18	79.25	80.80	61.17	62.23	63.35	54.79	56.80	56.53
	Claude-Sonnet-4.5	**82.01**	**84.50**	82.86	**67.19**	**70.11**	**67.47**	**61.78**	**65.42**	61.99
		Open-Source LLMs								
CoSIL	Qwen2.5-32B	30.77	19.34	83.50	–	–	–	22.11	14.85	55.38
Agentless		35.38	25.60	78.93	–	–	–	27.33	24.07	40.97
LocAgent		44.18	34.18	79.39	–	–	–	21.48	16.29	46.79
OrcaLoca		58.11	59.51	59.57	–	–	–	28.72	25.59	39.14
RepoSearcher	Qwen2.5-7B	30.09	18.80	83.11	–	–	–	25.57	17.68	62.38
	Qwen2.5-32B	32.25	20.24	**88.59**	–	–	–	28.03	19.36	**68.55**
RepoNavigator	Qwen2.5-7B	51.63	53.83	50.62	–	–	–	27.49	30.34	26.69
	Qwen2.5-14B	58.90	58.97	61.60	–	–	–	29.23	30.08	31.02
	Qwen2.5-32B	67.75	70.76	67.29	–	–	–	34.09	37.19	33.71
OpenHands-Bash	Qwen3-1.7B	2.40	2.09	3.60	0.37	0.32	0.60	0.34	0.32	0.50
	Qwen3-4B-Instruct	49.73	49.69	53.34	19.32	19.86	20.15	13.27	14.17	13.74
	Qwen3-14B	43.13	36.49	71.20	22.86	20.40	33.04	16.08	14.51	23.58
	Qwen3-32B (Thinking)	62.91	59.87	73.63	34.69	33.85	39.46	23.99	24.22	26.86
	CODESCOUT-4B	68.52	**71.53**	67.74	45.97	49.70	44.97	36.78	40.71	35.72
	CODESCOUT-14B	68.57	71.00	68.69	**50.88**	**53.71**	50.88	40.32	**43.74**	40.27
A²AGENT	Qwen3-4B-Instruct	72.64	70.88	76.76	46.94	42.25	59.55	38.59	35.50	48.98
	Qwen-8B	**73.31**	70.14	80.23	47.98	42.56	**63.42**	**41.01**	37.15	54.44

[1] Comparisons on SWE-Bench Verified

Method	LLM	File-level F1	File-level *P*	File-level *R*	Module-level F1	Module-level *P*	Module-level *R*	Function-level F1	Function-level *P*	Function-level *R*
		Closed-Source LLMs								
OpenHands-Bash	GPT-5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	Claude-Sonnet-4.5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
OpenHands-Bash^{rem}	GPT-5	61.18	69.10	62.06	42.20	52.86	39.63	35.86	48.91	32.65
	Claude-Sonnet-4.5	**64.75**	**75.39**	**64.49**	**48.54**	**61.75**	**45.39**	**42.26**	**58.14**	**38.06**
		Open-Source LLMs								
RepoSearcher	Qwen2.5-32B	3.81	2.52	9.00	–	–	–	2.31	2.46	2.52
LocAgent		19.77	0.38	25.73	–	–	–	4.30	0.17	8.72
Agentless		20.07	13.89	43.07	–	–	–	7.98	7.31	11.08
CoSIL		20.95	14.03	48.87	–	–	–	7.67	6.00	14.03
RepoNavigator	Qwen2.5-7B	39.74	48.13	36.36	–	–	–	14.29	21.26	12.33
	Qwen2.5-14B	49.72	58.64	46.85	–	–	–	18.06	25.25	16.05
	Qwen2.5-32B	57.57	68.69	53.49	–	–	–	20.72	29.44	18.13
OpenHands-Bash	Qwen3-1.7B	0.73	0.72	1.13	0.00	0.00	0.00	0.00	0.00	0.00
	Qwen3-4B-Instruct	36.96	44.42	35.59	11.78	17.46	10.19	8.12	12.16	7.01
	Qwen3-14B	30.08	28.48	48.22	11.87	13.82	14.21	8.20	9.97	9.92
	Qwen3-32B (Thinking)	46.85	49.65	54.55	21.94	27.18	22.62	12.31	17.82	11.93
	CODESCOUT-4B	51.77	68.98	46.16	36.97	**56.05**	31.13	29.03	**48.65**	23.73
	CODESCOUT-14B	53.63	68.81	48.81	37.13	53.80	32.02	28.74	46.09	23.76
A²AGENT	Qwen3-4B-Instruct	58.19	73.34	53.83	38.38	45.18	40.00	29.23	38.29	28.85
	Qwen3-8B	**60.05**	**76.54**	**56.16**	**39.07**	45.51	**42.37**	**30.60**	39.16	**30.91**

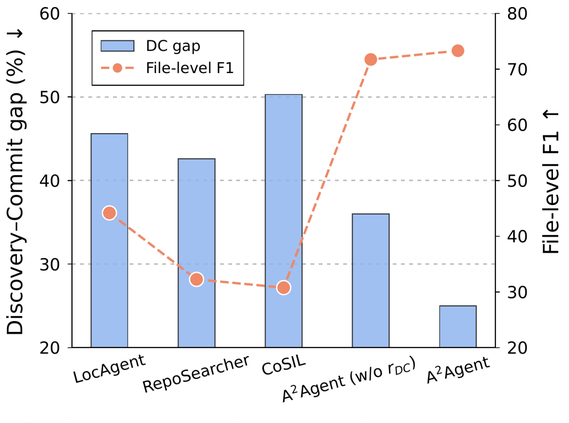
[2] Generalization on SWE-Bench Pro

Type	Method	Avg. #Turns (↓)	Avg. Time (↓)
Procedure-based	Agentless	2.77	6.41
Agent-based	CoSIL	17.55	77.57
	LocAgent	**6.20**	86.23
	RepoSearcher	12.47	42.58
	A²AGENT	11.72	**18.08**

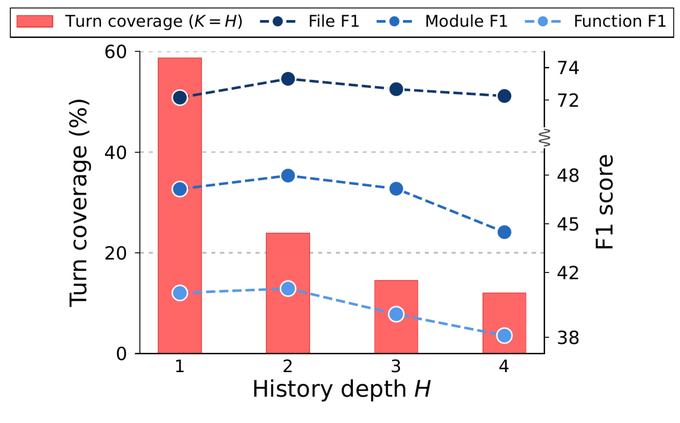
[3] Inference Efficiency

Method	File F1	Function F1	Resolved (%)
Oracle	-	-	45.93
RepoSearcher	29.33	18.02	26.82
CoSIL	32.70	19.37	27.64
Agentless	44.70	20.54	29.07
LocAgent	49.14	25.82	31.78
A²AGENT	**72.64**	**38.59**	**38.07**

[4] Issue Resolution Rate



[5] Discovery-Commit Gap Analysis



[6] Effect of History Depth H on F1 and Turn Coverage